

Getting the most from radiology AI: five things to measure

What a 4.5-year, 20-center study of multi-vendor AI in outpatient radiology means for heads of radiology, CMIOs and network managers

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WHAT WORKED, AND THE GAP

Over 4.5 years, a 20-center outpatient radiology network ran ten AI applications from seven vendors on about 389,000 examinations. Active adoption reached 91.4% of radiologists (53 of 58). Where AI was integrated into the reading workflow, radiologist-adjusted median report turnaround time was 26% lower for trauma radiography and 18% lower for knee MRI. At an annual platform cost equivalent to 23-25% of one radiologist FTE salary, trauma radiography alone corresponded to an estimated 0.69 FTE of annual reporting capacity (0.46 FTE radiologist-adjusted). The gap: 7.2% of AI results arrived after the report was finalized, and 72% of the median 2.06-minute latency was data routing rather than the algorithm.

91%

of radiologists adopted AI

-26%

adjusted report turnaround time, trauma radiography

80.6%

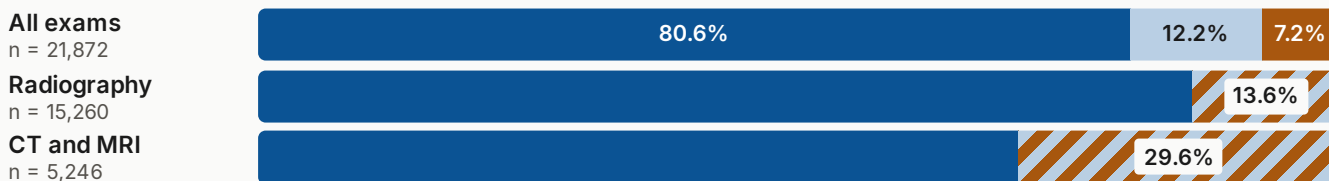
of AI results ready before the report was created; 7.2% arrived after finalization

72%

of latency was data routing, not the algorithm

WHEN THE AI RESULT ARRIVES

■ Before report creation ■ During reporting ■ After finalization (Too Late)
■ During or after finalization (group rows; reported combined)



What it means: most results are ready before reporting, but 7.2% arrive after finalization, and CT and MRI have the larger share arriving during or after reporting. Action: push routing and per-modality time-to-display targets, starting with CT and MRI. Source: Table 2 and Results text; mammography (n = 1,366) is in neither group.

WHAT NEEDS WORK, AND HOW TO FIX IT: FIVE THINGS TO MEASURE

- 1 Measure the share of AI results available before reporting, and the Too Late rate**

FINDING	80.6% of AI results were available before the report was created, but 7.2% arrived after the report was finalized (the Too Late rate).
WHAT IT MEANS	Those results came too late to change the report they belonged to.
ACTION	For every AI-processed examination, compare two HL7 timestamps: AI result available and report created and finalized. Track both shares monthly and bring the late share down.

2 Set a time-to-display target in contracts and acceptance tests

FINDING	CT and MRI results arrived during or after report finalization more often than radiography results (29.6% vs 13.6%).
WHAT IT MEANS	In cross-sectional imaging, the largest share of results still arrives while the radiologist is reporting or after.
ACTION	Agree, per modality, the maximum minutes from end of scan to AI result on the reading screen, and write it into the contract and the acceptance test (the paper's "Plan" step).

3 Prefer push routing and integration

FINDING	72% of the median 2.06-minute latency was data routing, not the algorithm. Fetching the study from PACS was the largest single step in five of nine solutions.
WHAT IT MEANS	Most of the delay sits in integration, so it can be reduced without changing the algorithm.
ACTION	Send studies to AI automatically at acquisition and return results into the reading tools (the paper's "Do" step); ask suppliers for routing and inference time separately.

4 Review latency stability every quarter

FINDING	Five of nine solutions had stable quarterly latency; four did not, showing progressive latency drift, post-deployment convergence, or a temporary infrastructure incident with full recovery.
WHAT IT MEANS	Latency changes after go-live and is invisible without a routine check.
ACTION	Plot quarterly median latency per solution with versioned metric definitions, review it with radiologist feedback, and put latency targets in contracts.

5 Survey your radiologists twice a year

FINDING	Perceived productivity stayed below neutral (2.94 of 5) despite measured time savings of 26% and 18%.
WHAT IT MEANS	Radiologists do not yet feel the gains that the timestamps show.
ACTION	Ask about usage, trust, quality, productivity (1 to 5) and a 0 to 10 recommendation score per tool; share the measured results, with training and regular feedback.

Regulation. The paper notes that all AI applications in the study fall under the EU AI Act (Regulation 2024/1689) high-risk classification, which mandates post-market monitoring aligned with these principles.

ABOUT THE EVIDENCE

- Observational and retrospective, in one private outpatient network in French-speaking Switzerland; other settings and architectures may differ.
- **Associations, not causation.** The comparison group without AI (2,429 of 20,909 examinations) was defined by consent refusal and technical failure, not by random assignment.
- **Timing, not use.** Results were displayed automatically, and the study did not record whether radiologists looked at them. The study measured latency, not algorithm accuracy.
- Survey analyses were exploratory, without multiplicity correction. The network has a royalty relationship with one AI supplier (knee MRI), disclosed in the paper.

Morozov S, Heracleous N, Korka D, Thouly C, Dufour B, Novarina O, Rizk B. AI Latency, Report Turnaround Time, and Adoption in a Multi-Vendor AI Ecosystem: A Multi-Site Observational Study. J Am Coll Radiol. 2026. doi:10.1016/j.jacr.2026.09.026
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